

ReliefLink - Git Commits Viva Defense Guide

Technical Rationale, Code Diff Analysis & Interview Explanations for the Recent 5 Commits

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💡 **Guide Pro-Tip for Git Inspection:** When the guide points to a commit on GitHub or in the terminal (`git log`), do NOT just say "I added 30 lines and deleted 20 lines." Explain **the business reason (why the code was modified)**, **the architectural tradeoff**, and **how the specific lines of code solve the problem**.

Commit 1: 943a7d8

new overall changes

Latest Commit • 7 files changed (+1305, -92)

1. CORE TECHNICAL PURPOSE

Eliminated the performance overhead of 3D WebGL globes on the landing page, tightened auth security by eliminating client-side OTP leaks, and introduced high-resolution Esri satellite GIS radars with real demo data.

2. KEY CODE CHANGES EXPLAINED

- **LandingPage.jsx:** Removed EarthGlobe (Three.js WebGL component) and view-toggle state (mapView). Replaced it with direct mounting of InteractiveMap (Leaflet satellite tiles). Added smooth translateY(-2px) CSS elevation to the Donate buttons.
- **LoginPage.jsx:** Removed the devOtp state and the yellow UI badge (`<div>Dev OTP: {devOtp}</div>`). The OTP is now printed **only to the backend terminal**.
- **SafetySatelliteMap.jsx:** Created a dedicated GIS satellite map component with Esri World Imagery + boundary overlays, live device GPS marker, and interactive pin-drop broadcast listeners.
- **seedKochiCamps.js:** Added a database seed script that populates 5 realistic active camps in Kochi/Ernakulam (Town Hall, Marine Drive, Kakkanad) into MongoDB Atlas.

3. WHAT TO SAY TO YOUR GUIDE

"In this commit, we tackled two critical concerns: **Performance** and **Security**. First, the 3D globe package was adding 1.86 MB of Three.js overhead that was impractical for local disaster response; we replaced it with high-resolution Esri Satellite imagery, which cut build times down to 2.9 seconds. Second, we removed the client-side dev OTP badge from LoginPage.jsx so authentication tokens are logged only in the protected server console, enforcing realistic production-grade security."

Commit 2: 3092552**Merge remote origin/main: keep satellite-only maps...**Merge Commit • 4 files changed
(+34, -42)**1. CORE TECHNICAL PURPOSE**

Standardized the mapping user experience across both public and responder views by removing confusing multiple map layers (OpenStreetMap normal, Esri Street, Humanitarian) and locking the map to **Satellite View only**.

2. KEY CODE CHANGES EXPLAINED

- **InteractiveMap.jsx & MapView.jsx:** Removed `<LayersControl>`, `<LayersControl.BaseLayer>`, and multiple alternate raster tile providers. Kept only Esri World Imagery (https://server.arcgisonline.com/.../World_Imagery/...) with city labels.
- **authController.js:** Streamlined user creation and OTP expiration handling so existing user profile names are preserved when returning victims log in via SMS.

3. WHAT TO SAY TO YOUR GUIDE

"In this merge commit, we resolved branch conflicts and eliminated visual clutter. The map previously had a radio-button layer picker with three alternate street maps that distracted from disaster coordination. We removed the extra tile layers and standardized on high-resolution satellite imagery so responders can instantly see geographical landmarks, rivers, and actual flooded terrain."

Commit 3: 3f1af58**enhance auth, payment gateway, NGO**Feature Commit • 26 files changed
(+3200, -733)**dashboard****1. CORE TECHNICAL PURPOSE**

Massive architectural expansion: Introduced a simulated financial payment gateway for monetary donations, upgraded the NGO donation verification pipeline, and implemented a responsive split-screen authentication portal.

2. KEY CODE CHANGES EXPLAINED

- **MockPaymentGateway.jsx & PublicDonatePage.jsx:** Built a realistic modal simulator supporting UPI QR/VPA, Credit/Debit Cards, and Net Banking with simulated bank processing delays and cryptographic transaction ID generation.
- **DonationsPage.jsx (NGO side):** Added complete administrative management for incoming aid: status filtering (pending, received, dispatched), monetary aggregation, donor phone verification, and receipt export.
- **LoginPage.jsx:** Re-engineered login architecture into a two-sided responsive split card: Left side dedicated to **Citizens (Phone + OTP)**, Right side dedicated to **Responders (Email + Password + JWT)**.
- **MissingPersonsPage.jsx:** Added photo uploads, last-seen location tags, and verification badges to assist in reuniting separated disaster victims.

3. WHAT TO SAY TO YOUR GUIDE

"Commit 3f1af58 was our core operational expansion. We built the complete end-to-end donation ecosystem. When a citizen makes a monetary donation, it flows through a simulated UPI/Card gateway, records the transaction payload in MongoDB, and instantly updates the NGO's camp finance dashboard. We also redesigned the login page to clearly distinguish evacuees from credentialed NGO/volunteer responders."

Commit 4: c095bc6**remove setup guide and boilerplate from**Cleanup Commit • 1 file changed (+0,
-55)**readme****1. CORE TECHNICAL PURPOSE**

Repository hygiene and documentation curation. Removed boilerplate instructions and scaffolding text to ensure professional academic presentation.

2. KEY CODE CHANGES EXPLAINED

- **README.md:** Stripped out default Vite/Create-React-App boilerplate instructions and local environment scratch notes, replacing them with project-focused architectural documentation.

3. WHAT TO SAY TO YOUR GUIDE

"This was a cleanup commit where we pruned default framework scaffolding and internal dev notes from the repository's root documentation to maintain a clean, production-standard GitHub repository for evaluation."

Commit 5: 59363b0**fix mobile network and non-gps relief**Algorithm & Fallback Commit • 6 files
changed (+488, -99)**camp discovery...****1. CORE TECHNICAL PURPOSE**

Resolved a critical real-world disaster scenario: **What happens when an evacuee's GPS signal is blocked by storm clouds or hardware failure?**

2. KEY CODE CHANGES EXPLAINED

- **locationStore.js**: Implemented a tiered geolocation strategy. If high-accuracy hardware GPS fails or times out, it falls back to cellular network/IP-based geolocation without failing the app.
- **CampFinderPage.jsx**: Added district-level and municipal keyword search filters, occupancy range sliders, and manual map-click override so users can discover relief shelters without hardware GPS.
- **campController.js**: Enhanced backend queries so the `/api/camps` endpoint dynamically switches between MongoDB `$nearSphere` geospatial distance queries (if coordinates exist) and indexed regex district filtering (if coordinates are missing).

3. WHAT TO SAY TO YOUR GUIDE

"Commit 59363b0 solves a life-critical edge case. In heavy rainfall or building collapse, satellite GPS signals frequently fail. Previously, the app required GPS to display nearby camps. In this commit, we built a multi-tier fallback: the system attempts GPS first; if that fails, it falls back to network geolocation, and simultaneously enables keyword district filtering so an evacuee can search 'Ernakulam' or 'Kochi' and instantly find open shelters with room available."